

## **Epic 4: Model Building**

### **Description**

This phase focuses on building, training, and evaluating machine learning models for the Credit Card Approval Prediction system. After the dataset has been cleaned and preprocessed, it is divided into training and testing sets to enable reliable model development and performance evaluation. Multiple classification algorithms are implemented and compared to determine which model provides the most accurate and consistent credit approval predictions.

Three supervised learning algorithms are used in this project: Logistic Regression, Random Forest, and Decision Tree. Each model is trained using applicant demographic information, financial details, and credit history features. The trained models generate predictions indicating whether a credit card application should be approved or rejected.

Model performance is evaluated using metrics such as Accuracy, Precision, Recall, F1-Score, and Confusion Matrix. Based on these evaluation results, the best-performing model is selected and saved for deployment in the web application.

### **Models Implemented**

- Logistic Regression Model
- Random Forest Model
- Decision Tree Model
- Performance comparison using evaluation metrics
- Selection and saving of the best-performing model

### **Outcome**

At the end of this phase, a trained and validated machine learning model is obtained for predicting credit card approval decisions. The selected model achieves reliable classification performance and is ready to be integrated into the Flask application for real-time prediction.